Summer intern 2026
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Report – GPS- Denied Navigation

This summer project investigated the use of a Kongsberg CP300 Acoustic Doppler Current Profiler (ADCP) and a Mini IMU for navigation of the Otter Uncrewed Surface Vehicle (USV).

The CP300, Wideband Transceiver (WBT), and the required power and network infrastructure were integrated onboard the vessel. EK80 bottom-track and water-track velocity measurements were collected with KM Binary navigation data and processed using a 15-state Error-State Kalman Filter (ESKF).

The main project deliverables were:

- Mechanical and electrical integration of the CP300 and WBT
- Network transfer and logging of KM Binary and EK80 VBW data
- Scripts for parsing, synchronizing and processing the recorded measurements
- Implementation of an Error-State Kalman Filter for GPS-denied navigation
- Calibration of EK80 velocity measurements
- Evaluation of depth cell size, measurement availability and lever-arm compensation

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1 System Overview

1.1 Maritim robotics

The vessel used in this project is an Otter Uncrewed Surface Vehicle (USV) supplied by Maritime Robotics. Throughout this report, the vessel is referred to as Gudrun.

Gudrun is operated remotely from a shore-based computer. An onboard computer located in the Otter's OBS box provides access to the vessel systems and is accessed remotely using Remmina. A 4G connection enables communication with the vessel and allows sensor data to be transferred to the development computer during testing.

The objective of this project is to estimate Gudrun's position when GPS measurements are unavailable. To achieve this, the vessel was equipped with a Mini IMU and a Kongsberg CP300 Acoustic Doppler Current Profiler (ADCP). The IMU provides measurements of the vessel's attitude and angular motion, while the CP300 measures the vessel's velocity relative to the seabed using bottom tracking. These measurements are fused in an Error-State Kalman Filter (ESKF) to estimate the position of the vessel during GPS-denied operation.

1.2 ADCP – cp300

The acoustic sensor used in this project is the Kongsberg CP300 Acoustic Doppler Current Profiler (ADCP). The CP300 is a compact Doppler sonar designed to measure both water velocity and vessel velocity relative to the seabed.

The transducer operates at a nominal frequency of 333 kHz within a frequency range of 270- 445 kHz. It uses four acoustic beams, each inclined 25° from the vertical, allowing velocity to be estimated in multiple directions. The CP300 supports both Continuous Wave (CW) and Frequency-Modulated (FM) pulse types.

The transducer has a depth rating of 1500 m, weighs approximately 2 kg in water, and supports selectable cell sizes of 0.5, 1, 2, 4, and 8 m.

The CP300 cannot operate independently. It must be connected to a Wideband Transceiver (WBT), which generates the acoustic pulses, receives the returning echoes, and transfers the processed measurements to the EK80 software for bottom-track and water-track processing.

1.3 System Integration

The CP300 sensor was mounted below the hull using a custom telescopic mounting system to ensure that the transducer remained fully submerged during operation while allowing the installation height to be adjusted when required.

The Wideband Transceiver (WBT) was installed inside a waterproof enclosure mounted on the vessel. All Ethernet and power connections were routed and secured to protect them from water ingress and mechanical damage during field testing.

The original plan was to power the WBT from the regulated power output available on the PCB inside the Otter's OBS box. During testing, this supply proved insufficient to provide stable operation of the acoustic system while the vessel's onboard electronics were running. A dedicated 12 V battery was therefore installed to power the WBT independently, providing a stable and reliable power supply throughout all experiments.

The CP300 was removed from the vessel between test campaigns and reinstalled before each deployment. Although the mounting system was designed to reproduce the sensor position, small rotational changes could occur when the telescopic bracket was reassembled. Consequently, the alignment between the ADCP frame and the vessel body frame could not be assumed to remain identical between installations. A new velocity alignment calibration should therefore be performed after each reinstallation.

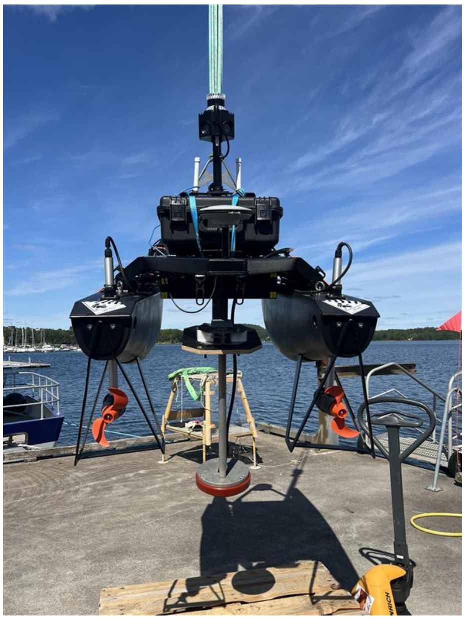

Figure 1-CP300 telescpoic mounting

2 Data Logging

2.1 IMU log (KM Binary)

The navigation data were transmitted as Kongsberg KM Binary (KMB) messages at approximately 100 Hz. Each message contains the vessel's position, orientation, angular rates, velocity and acceleration.

The GPS position was used only as reference data during calibration and evaluation of the navigation filter. After the filter was initialized, the GPS measurements were not used to correct the estimated trajectory.

The most important fields used in this project are listed in Table 1-IMU

latitude_deg, longitude_deg	degrees	Reference trajectory
roll_deg, pitch_deg, heading_deg	degrees	Vessel orientation
rollRate, pitchRate, yawRate	deg/s	Rotational motion
velNorth, velEast, velDown	m/s	Calibration reference
northAcceleration, eastAcceleration, downAcceleration	m/s^2	Filter prediction

Table 1-IMU

2.2 EK80

The CP300 ADCP provides vessel velocity measurements through the EK80 software using the NMEA 0183 VBW message format. Each message contains both water-referenced and bottom-referenced velocity components.

water_forward	Forward velocity relative to the water
water_transverse	Transverse velocity relative to the water
bottom_forward	Forward velocity relative to the seabed
bottom_transverse	Transverse velocity relative to the seabed

Table 2 - WBV

The runtime filter configuration used bottom-track velocity as the primary measurement source, automatically switching to water-track velocity whenever bottom-track was unavailable. Section 7.2 shows that this fit was better for the water-track channel, and Sections 7.3 and 8 show that this hybrid, bottom-first configuration is not actually the best-performing option across all collected data.

The navigation and EK80 data were collected using separate logging methods. The EK80 software was executed on the Windows payload computer inside the Otter's OBS box, where the built-in UDP output function was used to transmit VBW messages over the network.

The IMU navigation data (KM Binary) were forwarded from the payload computer to the development computer using a custom Python script. Both data streams were transferred through the Tailscale network, allowing real-time access from the development computer during field testing.

The IMU and EK80 measurements were logged separately as CSV files and synchronized afterwards using timestamps. This enabled comparison with the GPS reference trajectory and calibration of the EK80 velocity measurements before they were used by the Error-State Kalman Filter.

3 Measurement Uncertainty

During the field tests, the EK80 occasionally lost bottom lock, resulting in missing or invalid bottom- track measurements. In addition, individual measurements may contain outliers caused by acoustic disturbances or poor bottom tracking.

Short interruptions in the bottom-track data are expected during normal operation. Instead of immediately terminating the navigation solution, the filter handles missing measurements according to the duration of the outage.

- **$\Delta t \leq 5$ s:** The filter continues normal dead reckoning using the latest IMU measurements.
- **5 s $< \Delta t \leq 120$ s:** Position, velocity and orientation are no longer updated by velocity measurements. The covariance matrix is increased to reflect the growing uncertainty.
- **$\Delta t > 120$ s:** The outage is considered too long for reliable navigation, and the measurement update is skipped until valid data become available again.

4 Error state Kalman filter

4.1 state vector

The nominal state consists of the vessel position, velocity, orientation and sensor biases

Symbol	Description	Unit
p	Position, local NED frame	m
v	Velocity, local NED frame	m/s
q	Orientation, body → NED	unit quaternion
b_a	Accelerometer bias	m/s^2
b_g	Gyroscope bias	rad/s

Table 3 -ESKF nominal state

The corresponding state is :

$$\mathbf{x} = [\mathbf{p} \;\; \mathbf{v} \;\; \mathbf{q} \;\; \mathbf{b}_a \;\; \mathbf{b}_g]^\top \in \mathbb{R}^{16}$$

The corresponding error state is :

$$\delta\mathbf{x} = [\delta\mathbf{p} \;\; \delta\mathbf{v} \;\; \delta\boldsymbol{\theta} \;\; \delta\mathbf{b}_a \;\; \delta\mathbf{b}_g]^\top \in \mathbb{R}^{15}$$

Nominal state vector x (top) and error-state vector δx (bottom).

4.2 Coordinate system

The navigation filter operates in a local North-East-Down (NED) coordinate frame. GPS latitude and longitude are converted into local Cartesian coordinates using the vessel's initial position as the reference point:

$$N = (\varphi - \varphi_0)\, R_{\oplus}$$

$$E = (\lambda - \lambda_0)\, R_{\oplus} \cos\varphi_0$$

Equation 1- lat/lon to local NED

This transformation allows all calculations to be performed in meters instead of geographical coordinates.

4.3 Orientation representation

The vessel orientation is represented by a unit quaternion

$$q = [w, x, y, z].$$

$$\| q \| = 1.$$

Quaternions are preferred over Euler angles because they avoid singularities, provide improved numerical stability and are computationally efficient when combining rotations.

Two operations are used throughout the filter: quaternion multiplication, used to compose rotations

$$q_1 \otimes q_2 = \begin{bmatrix} w_1 w_2 - x_1 x_2 - y_1 y_2 - z_1 z_2 \\ w_1 x_2 + x_1 w_2 + y_1 z_2 - z_1 y_2 \\ w_1 y_2 - x_1 z_2 + y_1 w_2 + z_1 x_2 \\ w_1 z_2 + x_1 y_2 - y_1 x_2 + z_1 w_2 \end{bmatrix}$$

Equation 2- $q_1 \otimes q_2$ quaternion (Hamilton) product.

and the corresponding rotation matrix, used whenever a vector needs to be rotated between the body frame and the NED frame:

$$R(q) = \begin{bmatrix} 1 - 2(y^2 + z^2) & 2(xy - zw) & 2(xz + yw) \\ 2(xy + zw) & 1 - 2(x^2 + z^2) & 2(yz - xw) \\ 2(xz - yw) & 2(yz + xw) & 1 - 2(x^2 + y^2) \end{bmatrix}$$

Equation 3- R(q), the rotation matrix equivalent of the quaternion.

To propagate the orientation forward by one small time step Δt, the quaternion is updated using the measured angular velocity, corrected for the estimated gyroscope bias

$$\delta \mathbf{q} = \left[\cos \frac{\|\boldsymbol{\omega} \Delta t\|}{2}, \quad \frac{\boldsymbol{\omega} \Delta t}{\|\boldsymbol{\omega} \Delta t\|} \sin \frac{\|\boldsymbol{\omega} \Delta t\|}{2} \right]$$

Equation 4- δq, small-angle quaternion increment over Δt.

4.4 Prediction step

The prediction step is executed for every IMU measurement. The corrected IMU measurements are:

$$\omega = \omega_m - b_g$$
$$a = a_m - b_a$$

Where:

- $\omega_m$is the measured angular velocity,
- $a_m$is the measured acceleration.

The nominal state is propagated using standard strapdown integration (position, velocity and orientation updated from the corrected IMU measurements). The error-state covariance matrix is propagated using:

$$P_{k+1} = F P_k F^T + Q$$

Equation 5

where F is the discrete state-transition matrix, Q is the process noise covariance, accounting for IMU measurement noise and slowly varying sensor biases (modelled as random walks).

$$F = \begin{bmatrix} I & I\Delta t & 0 & 0 & 0 \\ 0 & I & 0 & -I\Delta t & 0 \\ 0 & 0 & I - [\omega\times]\Delta t & 0 & -I\Delta t \\ 0 & 0 & 0 & I & 0 \\ 0 & 0 & 0 & 0 & I \end{bmatrix} \quad [\omega\times] = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

Equation 6 -State-transition matrix F (left) and process-noise covariance Q. [ω×] is the skew-symmetric matrix used to linearise the effect of a rotation rate on the orientation error.

4.5 Measurement update

Whenever a valid EK80 velocity measurement becomes available, the predicted velocity is compared with the measured velocity. Conceptually, the filter is combining two independent estimates of the same quantity , a prediction from the motion model and a measurement from the sensor, each with its own uncertainty (σ). It weighs each one inversely to its uncertainty: the narrower (more confident) distribution pulls the combined estimate closer to itself. This weighting is exactly what the Kalman gain computes, and the combined estimate is always at least as certain as either input alone:

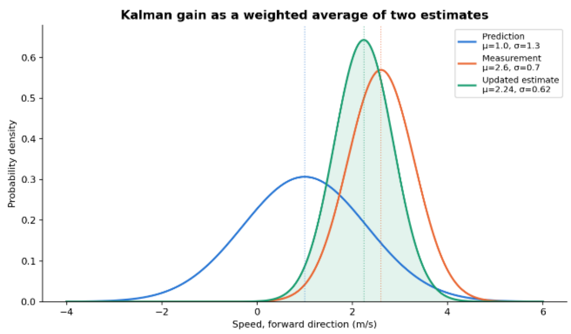

Figure 2 - The Kalman gain as a weighted average of two Gaussian estimates. Combining two independent measurements can only reduce uncertainty, never increase it.

The innovation (the discrepancy between predicted and measured velocity)is computed as:

$$y = z - h(x).$$

The Kalman gain is then calculated as

$$K = PH^T(HPH^T + R)^{-1}.$$

Equation 7- kalman gain

The estimated error state is injected into the nominal state, and the error state is reset.

$$\mathbf{p} \leftarrow \mathbf{p} + \delta\mathbf{p} \qquad \mathbf{v} \leftarrow \mathbf{v} + \delta\mathbf{v}$$

$$\mathbf{q} \leftarrow \mathbf{q} \otimes \exp(\tfrac{1}{2}\delta\boldsymbol{\theta})$$

$$\mathbf{b}_a \leftarrow \mathbf{b}_a + \delta\mathbf{b}_a \qquad \mathbf{b}_g \leftarrow \mathbf{b}_g + \delta\mathbf{b}_g$$

$\delta\mathbf{x} \leftarrow 0$ (feiltilstanden nullstilles etter injeksjon)

Equation 8 - Error-state injection and reset.

4.6 Mahalanobis distance

When a bottom-track measurement is available, it is not necessarily correct. Before each measurement is used by the filter, the innovation is evaluated using the Mahalanobis distance.

$$d_M^2 = \mathbf{y}^\top \mathbf{S}^{-1}\mathbf{y}$$

$\mathbf{y} = \mathbf{z} - \hat{\mathbf{z}}$ (innovasjon: målt minus forventet fart)

$\mathbf{S} = \mathbf{HPH}^\top + \mathbf{R}$ (innovasjonens kovarians)

Avvis målingen hvis $d_M^2 >$ terskel

Equation 9- Mahalanobis distance, where y is the innovation and S is the innovation covariance matrix.

where y is the innovation and S is the innovation covariance matrix.

Unlike a fixed velocity threshold, the Mahalanobis distance takes the filter's estimated uncertainty into account. Immediately after a valid measurement, the filter has low uncertainty and therefore rejects even relatively small deviations. After a long measurement outage, the uncertainty is larger, allowing larger innovations to be accepted.

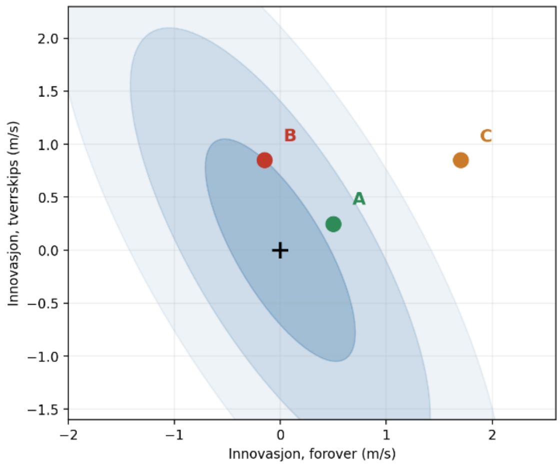

Figure 3- Three candidate innovations (A, B, C) against the filter's current uncertainty ellipse. A and B fall inside the gate and are accepted; C is rejected as an outlier despite having a similar forward-direction offset to B, because the ellipse's orientation

4.7 Velocity calibration

The raw EK80 velocity measurements cannot be used directly, because they contain systematic scale errors and offsets relative to true vessel speed. A linear least-squares model is therefore fitted between the raw EK80 velocity components and a reference velocity obtained from the navigation system (GPS position and INS attitude, rotated into the vessel's body frame):

$$v^{true} \approx \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \end{bmatrix}$$

Equation 10 - Linear calibration model for the forward/transverse velocity channels.

Separate models of this form are fitted independently for the forward and transverse channels, using ordinary least squares:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 \quad \Rightarrow \quad \hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$$

with goodness of fit assessed using the coefficient of determination:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

Equation 11 - Ordinary least-squares fit (left) and coefficient of determination R

Fitted result (27.07 dataset, water-track channel, forward speed): $c_{11} = 0.143$, $c_{12} = -0.445$, $c_{13} = 0.086$, giving $R^2 = 0.814$ over N = 2,913 points, with a residual noise (1σ) of 0.124 m/s. An R^2 of 0.81 means the calibrated model explains about 81% of the variance in true forward speed; the remaining scatter is residual sensor noise plus timing jitter between the EK80 and the navigation system.

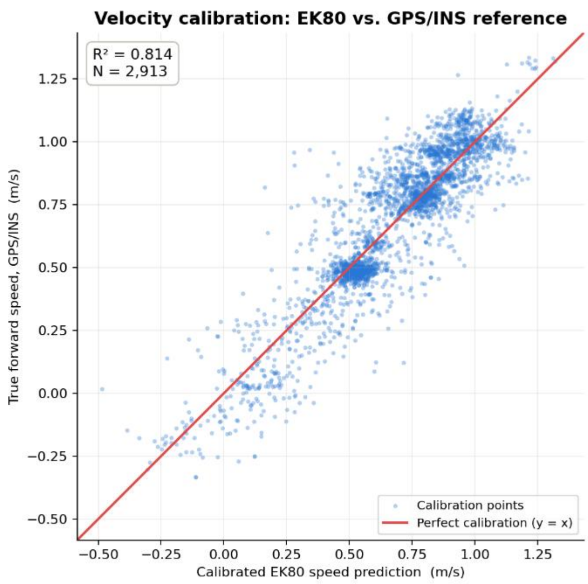

Figure 4 -Velocity calibration of EK80 water-track measurements against the GPS/INS reference for the 27 July dataset. The red line indicates perfect agreement.

The EK80 velocity measurements were calibrated against the GPS/INS reference using a linear calibration model. For the 27 July dataset, the calibrated water-track measurements achieved an R^2 of 0.81, demonstrating good agreement with the reference trajectory. This calibration model was used in the subsequent filter evaluation.

4.8 Calibration Repeatability

Although the calibration shown in Figure 5-provides a good fit for the 27 July dataset, additional analysis revealed that the calibration is not stable across separate test campaigns.

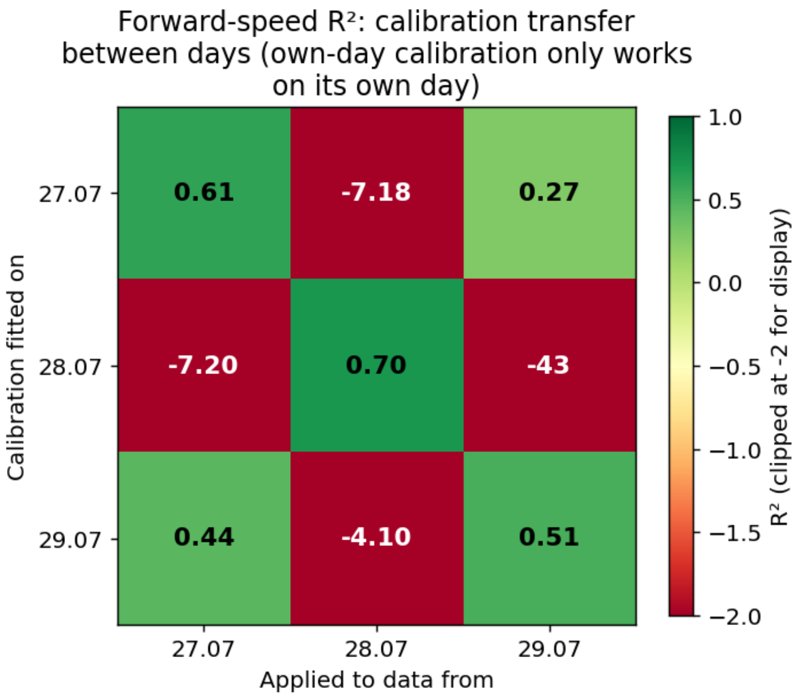

Figure 5- Cross-day validation of the EK80 velocity calibration. Each calibration model performs well on the dataset from the same day (diagonal), but performs poorly when applied to data collected after a separate installation of the CP300.

These results indicate that the calibration is sensitive to small changes in sensor alignment after reinstallation of the CP300. Consequently, a new calibration should be performed after each reinstallation.

4.9 Uncertainty estimation

The covariance matrix represents the filter's estimated uncertainty. The horizontal position covariance is converted into a 3σ confidence margin to visualize how the expected position uncertainty develops over time.

$$\sigma_N = \sqrt{P_{11}}, \qquad \sigma_E = \sqrt{P_{22}}$$

$$\text{Margin} = n_\sigma \cdot \max(\sigma_N,\ \sigma_E) \qquad (n_\sigma = 3 \text{ som standard})$$

Equation 12 - 3σ confidence radius from the horizontal position covariance.

The 3σ margin was selected empirically and should be interpreted as an indication of relative uncertainty rather than as a strict statistical confidence bound. In particular, the covariance model does not explicitly account for ocean current when water-track measurements are used.

4.10 Performance evaluation

To evaluate the effect of the EK80 velocity measurements, each dataset was processed twice using the same initial conditions. In the first case, the filter operated in open-loop, relying solely on IMU prediction after initialization. Since no external measurements were used to correct the estimated state, the position error gradually increased due to the accumulation of IMU errors.

In the second case, the filter operated in closed loop, using valid EK80 velocity measurements. Bottom-track was used as the primary measurement source, while water-track was used as fallback whenever bottom-track was unavailable. Each accepted velocity measurement corrected the predicted state and reduced the accumulated drift.

The estimated trajectories from both configurations were compared with the GPS reference. The position error was calculated after each measurement update, and the root mean square error (RMSE) was used to quantify the improvement obtained by incorporating the EK80 velocity measurements.

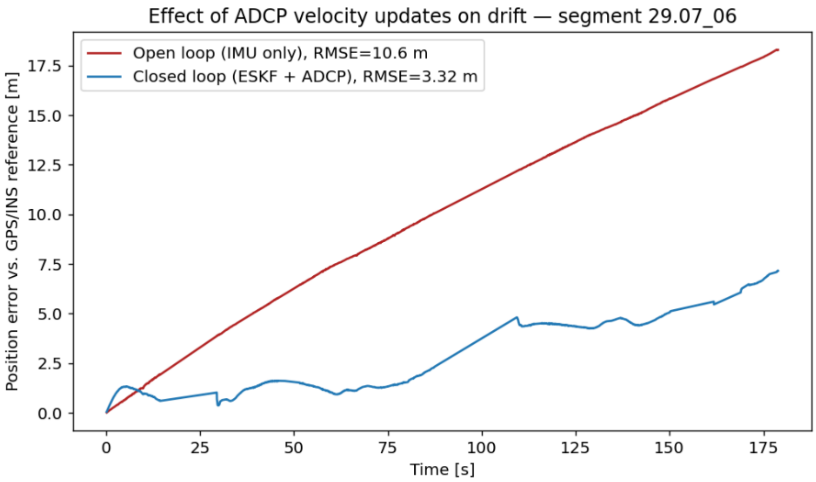

Figure 6 -Position error for segment 02 (28 July). Without EK80 velocity updates (open-loop, red), the position error increases continuously due to inertial drift. With EK80 updates enabled (closed-loop, blue), the position error remains bounded relative to the GPS/INS reference.

4.11 Results

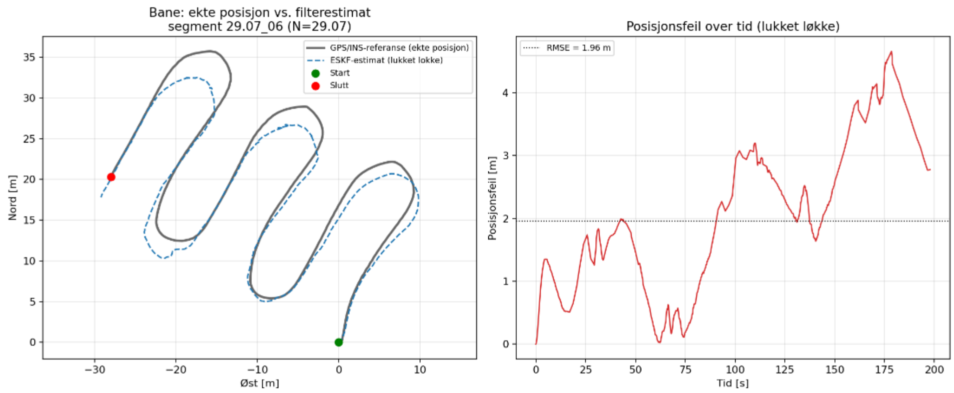

Figure 7 - Estimated trajectory (left) compared with the GPS/INS reference trajectory and the corresponding horizontal position error (right) for one representative experiment (segment 29.07_06, RMSE = 1.96 m).

The estimated trajectory follows the reference trajectory closely, with only small deviations during the run. The position error remains bounded and does not exhibit continuous drift. For the example shown, the root mean square error (RMSE) is 1.96 m.

The same evaluation procedure was applied to all test segments. Although the RMSE varied between experiments, the filter consistently maintained a stable navigation solution when valid EK80 velocity measurements were available.

4.12 Evaluation Measure: NIS

In addition to the RMSE, the filter consistency was evaluated using the Normalised Innovation Squared (NIS),

$$\text{NIS} = \mathrm{y}^T S^{-1} \mathrm{y},$$

Equation 13 -NIS

where y is the innovation and S is the innovation covariance. For a consistent filter with two measurement dimensions, the NIS is expected to follow a χ^2distribution with two degrees of freedom.

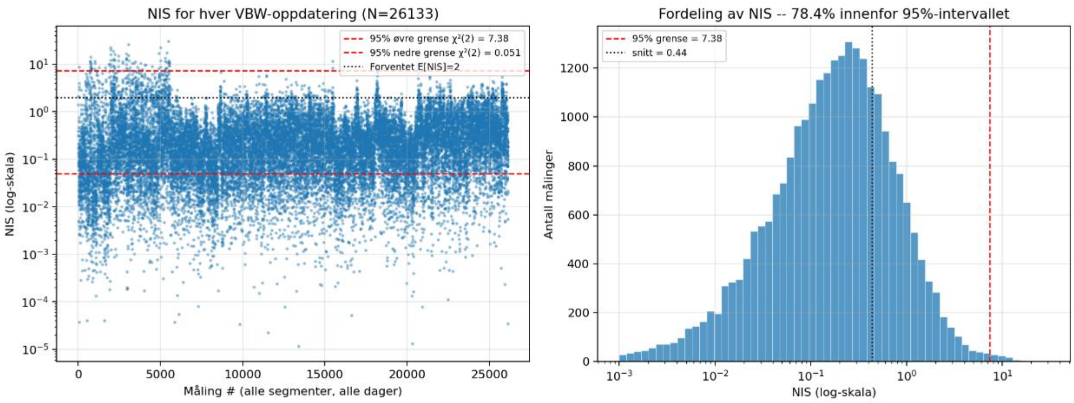

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Figure 8 - Normalised Innovation Squared (NIS) for all accepted EK80 velocity updates. The left panel shows the NIS values for each accepted measurement, while the right panel shows the corresponding histogram. The dashed red lines indicate the 95% confidence bounds for a χ^2 distribution with two degrees of freedom.

The mean NIS was 0.44. Approximately 78.4% of the measurements were within the theoretical 95% interval, while only 0.4% exceeded the upper bound. The low mean NIS indicates that the filter is conservatively tuned, suggesting that the assumed measurement uncertainty may be larger than necessary

To confirm that this behavior is representative and not specific to the segment shown in Figure 8, the same NIS analysis was repeated across all 34 recorded segments. Aggregated over 47,848 accepted updates, the mean NIS was 0.60 and 99.5% of updates fell within the 95% confidence bound, confirming the same conservative-but-consistent filter behavior across the full dataset rather than in a single example.

5. Arm

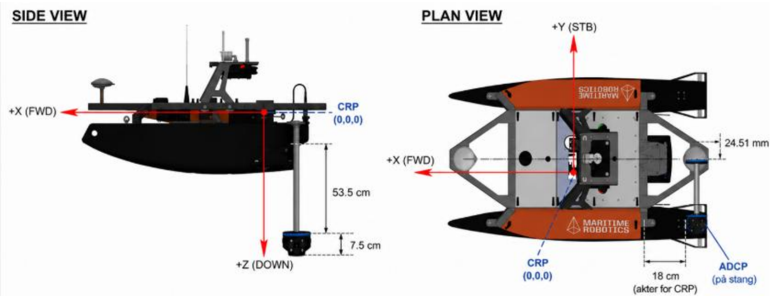


Figure 9 – Otter/Gudrun

The lever-arm distance was approximately fixed by the mechanical design, while the rotational alignment was more sensitive to reinstallation.

5.1 Lever-arm between the IMU and ADCP

The CP300 ADCP is mounted at an offset from the IMU, which defines the Common Reference Point (CRP) used by the Error-State Kalman Filter. The lever arm, expressed in the vessel body frame, is

$$\mathbf{r}_{ADCP} = [-0.18,\ 0.245,\ 0.610]\ \mathrm{m}.$$

Since the navigation state is estimated at the CRP while the velocity is measured at the ADCP, the measured velocity contains an additional rotational component caused by the vessel's angular motion. This component is given by

$$\mathbf{v}_{LA} = \boldsymbol{\omega} \times \mathbf{r}$$

Equation 14 -arm

where $\boldsymbol{\omega}$is the angular velocity measured by the IMU.

During straight-line motion the correction is small, whereas it increases during turning manoeuvres as the angular velocity increases.

5.2 Lever-arm correction

The lever-arm correction was applied to every EK80 velocity measurement before the measurement update in the Error-State Kalman Filter.

$$\mathbf{z}_{corr} = \mathbf{z} - (\boldsymbol{\omega} \times \mathbf{r})_{xy}$$

Equation 15- Lever-arm

where only the forward and transverse velocity components are used by the filter.

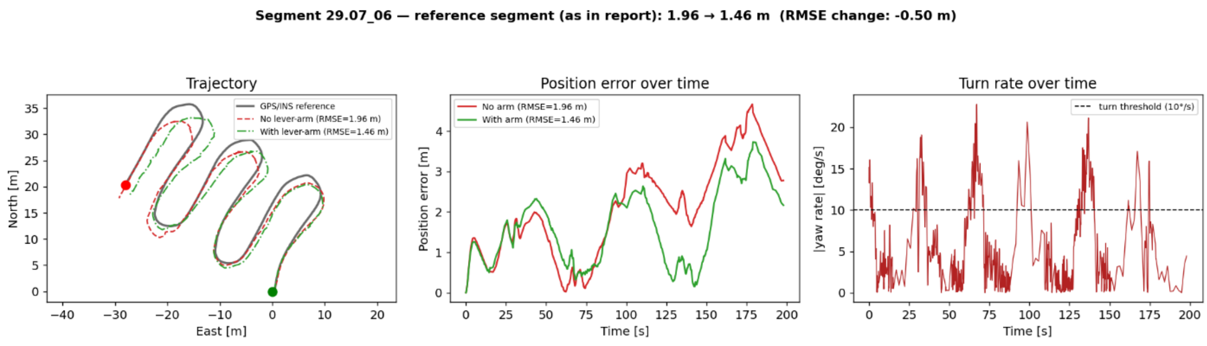


Figure 10 - Representative test segment containing several turning manoeuvres. Applying the lever-arm correction reduced the RMSE from 1.96 m to 1.46 m. to an improvement of approximately 26%.

During straight-line motion, the angular velocity is close to zero, making the rotational component negligible. Consequently, the lever-arm correction has little influence on the estimated trajectory during steady motion.

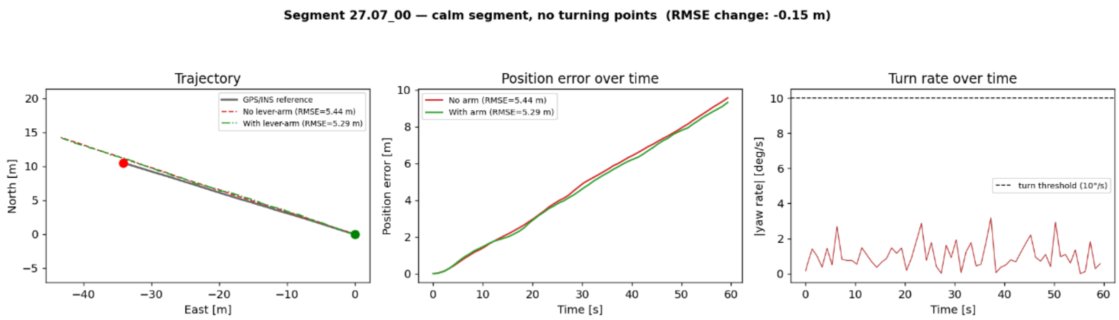
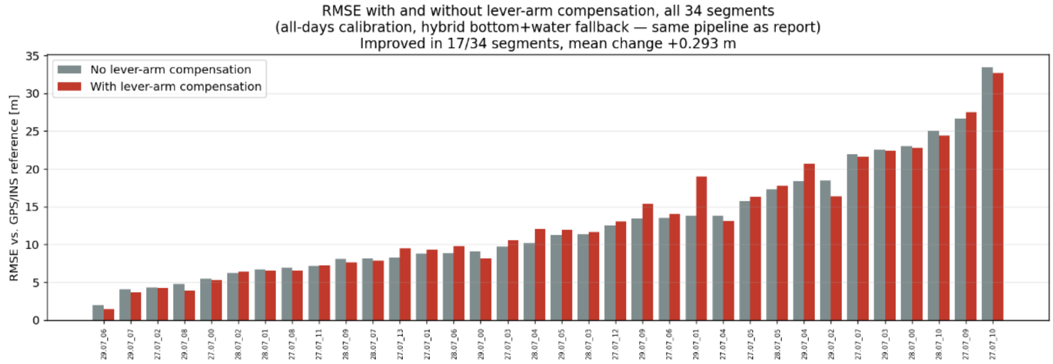


Figure 11-Example of a segment dominated by straight-line motion. Because the vessel experiences little angular motion, the lever-arm correction has only a negligible effect on the estimated trajectory and RMSE.

The contrast between Figures 10 and 11 illustrates that the effectiveness of the lever-arm correction depends primarily on the vessel's angular velocity. Segments containing multiple turning manoeuvres benefit from the correction, whereas segments dominated by straight-line motion show little or no improvement.



The correction improved the whole-segment RMSE in 17 of the 34 evaluated segments. The mean RMSE change across all segments was +0.29 m, indicating that the overall influence on whole-segment accuracy is small.

This does not contradict the results shown in Figure 10. Since most samples in a typical trajectory are collected during straight-line motion, the localized improvements obtained during turning manoeuvres contribute only modestly to the overall segment RMSE.

6. EK80 parameter Evolution

The performance of the EK80 bottom-track measurements depends on the selected acquisition parameters. In this project, different depth cell sizes were evaluated to determine which configuration provides the best balance between measurement noise and measurement availability for GPS-denied navigation.

6.1 Evaluation of Depth Cell Size

To investigate how the selected depth cell size influences the quality of the EK80 bottom-track measurements, repeated field tests were carried out using depth cell sizes of 1 m, 2 m and 4 m. The objective was to evaluate both the availability of valid bottom-track measurements and the measurement noise of the corresponding velocity estimates.

An initial test with an 8 m depth cell was performed on 28 July inside the harbour, where the average water depth was approximately 2 m. Although an 8 m cell size is recommended by the manufacturer under suitable conditions, the EK80 was unable to obtain bottom lock in such shallow water. Consequently, almost no valid bottom-track measurements were obtained, and this configuration was excluded from further analysis.

The remaining experiments were conducted using 1 m, 2 m and 4 m depth cells.

Test	Date	Area	Mean depth	Ping interval	cell sizes
Serie A	28.07	Inside the harbour	2	1,0 s	1, 2, 4 (+ 8)
Serie B	29.07	Outside the harbour	11 m	max (0,04 s)	1, 2, 4
Serie C	29.07	Inside the harbour	2 m	max (0,04 s)	1, 2, 4

Table 4 -Depth-cell test series and conditions.

The ping interval was changed between the experiments performed on 28 July and 29 July. Since the ping rate affects the number of measurements, the temporal resolution and the estimated noise metrics, datasets collected on different days cannot be compared directly. Comparisons are therefore made only between depth cell sizes within the same test series.

During several experiments, the planned route had to be terminated before completion due to stability on the Otter. To ensure a fair comparison, all datasets were trimmed to the common section of the route completed in every test. The overlapping section was identified using the Seapath GPS trajectory.

The experiments conducted on 29 July consisted of two different environments:

Series B: Deep-water area outside the harbour (average depth approximately 11 m).

Series C: Shallow-water area inside the harbour (average depth approximately 2 m).

6.2 Performance Metrics

Two complementary noise metrics were used: the first-difference method (σ_{diff}) and the Savitzky–Golay residual method (σ_{savgol}). Their definitions are presented in Section X. In addition, measurement availability was evaluated using the percentage of invalid bottom-track measurements, the number of dropout periods and the longest continuous dropout.

The first-difference method estimates the short-term random variation by calculating

$$\sigma_{\text{diff}} = \frac{\text{std}(\Delta v)}{\sqrt{2}}$$

Equation 16 - first-difference method

where $\Delta v$is the difference between consecutive velocity measurements.

The second method estimates the noise by subtracting a Savitzky–Golay smoothed signal from the measured velocity,

$$\sigma_{\mathrm{savgol}} = \mathrm{std}(v - v_{\mathrm{smooth}}).$$

Equation 17-Savitzky–Golay

6.3 Results

Tables 5 -7 summaries the results obtained for the three depth cell sizes.

Overall, increasing the depth cell size reduced the estimated noise measurement among the valid measurements. This improvement was accompanied by a reduction in bottom-track availability, particularly for the 4 m configuration.

6.3.1 Serie A 28.07 (ping 1,0 s)

Cell size	σ_diff, forover	σ_diff, tvers	σ_savgol, forover	σ_savgol, tvers
1	0,146	0,118	0,088	0,083
2	0,077	0,068	0,045	0,050
4	0,072	0,048	0,028	0,027

Table 5 -Series A noise metrics.

The results show that both $\sigma_{\mathrm{diff}}$and σ_{savgol}decrease as the depth cell size increases. Among the valid measurements, the 4 m configuration produced the lowest estimated noise.

6.3.2 Serie B -29.07, outside harbour (ping max, 102 m)

Cell size	Pings	Invalid bottom-track	Drop-periodes	Longest dropout	σ_diff (fwd/tvers)	σ_savgol (fwd/tvers)
1	2 932	53 (1,8 %)	26	4 pings	0,154 / 0,161	0,111 / 0,117
2	3 168	12 (0,4 %)	10	2 pings	0,064 / 0,059	0,044 / 0,039
4	1 349	817 (60,6 %)	66	34 pings	0,027 / 0,036	0,018 / 0,024

Table 6- Series B noise metrics.

The 4 m depth cell produced the lowest estimated noise, but more than 60% of the bottom-track measurements were invalid. In contrast, the 2 m configuration provided nearly continuous bottom tracking, with only 0.4% invalid measurements and a maximum dropout of only two consecutive pings. Although the 1 m configuration also maintained good availability, its estimated noise was considerably higher.

6.3.3 Serie B 29.07, inside harbour (ping max)

File	Cell size	Pings	Invalid bottom-track	Drop-periodes	ongest dropout	σ_diff (fwd/tvers)	σ_savgol (fwd/tvers)
08	1	4 484	15 (0,3 %)	12	3 pings	0,124 / 0,121	0,089 / 0,085
07	2	4 207	30 (0,7 %)	10	8 pings	0,051 / 0,047	0,033 / 0,031
06	4	1 711	346 (20,2 %)	53	23 pings	0,041 / 0,035	0,020 / 0,019

Table 7- Series C noise metrics.

A similar trend was observed in the shallow-water environment. The 4 m configuration again produced the lowest estimated noise but experienced substantially more bottom-track dropouts than the 1 m and 2 m configurations. The 2 m configuration achieved considerably lower noise than the 1 m configuration while maintaining high measurement availability.

6.4 Discussion

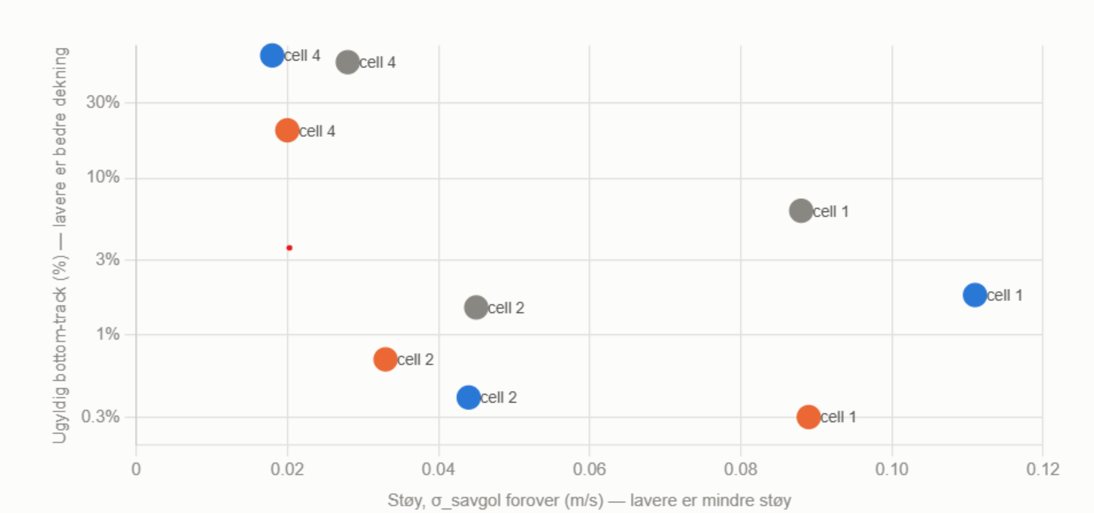


Figure 12 - blue, orange and grey markers represent Series A, B and C, respectively, while each point is labelled with its associated depth cell size. The optimal operating point lies towards the lower-left corner: low noise and high availability.

The results demonstrate a clear trade-off between measurement quality and bottom-track availability. Increasing the depth cell size consistently reduced both $\sigma_{\mathrm{diff}}$and $\sigma_{\mathrm{savgol}}$for the valid measurements.

The lower noise observed for the 4 m configuration does not necessarily indicate superior overall performance. The noise analysis only includes measurements for which the EK80 successfully maintained bottom lock. If weaker or less reliable measurements are rejected by the EK80, the remaining dataset will naturally appear less noisy. This introduces a selection effect that should be considered when interpreting the results.

For the Error-State Kalman Filter, continuous velocity updates are essential. During periods without bottom-track measurements, the navigation solution relies solely on IMU propagation, causing the position uncertainty to increase over time. Consequently, a configuration with very low measurement noise but frequent measurement outages may provide poorer overall navigation performance than a configuration with slightly higher noise but significantly better measurement availability.

Overall, the 2 m depth cell provided the best compromise between measurement noise and bottom-track availability. Compared with the 1 m configuration, it achieved substantially lower noise while maintaining nearly continuous bottom tracking. Although the 4 m configuration produced the lowest noise among valid measurements, its significantly higher dropout rate makes it less suitable for continuous GPS-denied navigation.

7. Comparison of Bottom-Track and Water-Track Measurements

In some areas the bottom-track (DVL-like) signal drops out or loses lock, while the water-velocity channel from the same EK80 log remains available throughout. This section quantifies how often that happens across all three collection days and evaluates whether water velocity should be used more than as a pure fallback.

7.1 Data availability

Water velocity was available for 100% of all logged measurements throughout the three test days (52,293 samples). Bottom-track availability varied between 71.5% and 93.6%, with two complete test segments on 27 July containing no valid bottom-track measurements.

Day	Water coverage	Bottom coverage
27.07	100%	71.5%
28.07	100%	91.9%
29.07	**100%**	**93.6%**

Table 8- Per-day data coverage.

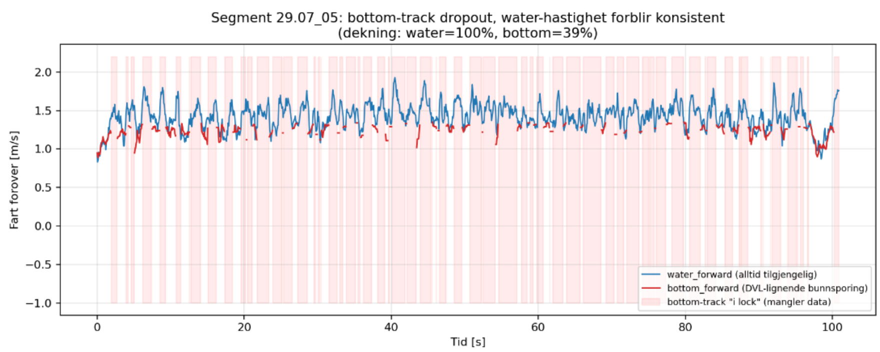


Figure 13- a typical bottom-track dropout. While bottom-track was available for only 39% of the segment, the water velocity remained continuous throughout the recording.

7.2 Measurement comparison

Both velocity measurements were compared with the GPS/INS reference.

Day	Water R^2	Bottom R^2
27.07	0.81	0.61
28.07	0.66	0.70
29.07	0.57	0.51

Table 9 -Per-day calibration quality (R^2 against GPS/INS reference, forward channel).

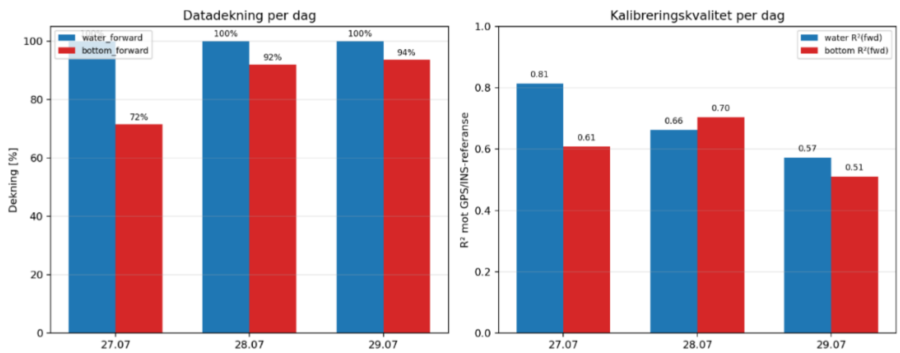


Figure 14 -Water-track vs. bottom-track velocity, all days pooled where both channels are valid simultaneously (r = 0.84, mean difference 0.02 m/s).

Water velocity achieved a higher correlation than bottom-track on two of the three test days. When both measurements were available simultaneously, they showed a correlation coefficient of r = 0.84 with a mean difference of -0.02 m/s, indicating no significant systematic bias.

7.3 Navigation performance

The closed-loop ESKF was evaluated using three measurement configurations: water-only, bottom-only, and the current hybrid solution with bottom-track as the primary measurement and water velocity as fallback.

Configuration	Mean RMSE	Median RMSE
Water only	12.43 m	9.88 m
Hybrid	12.68 m	10.72 m
Bottom only	12.86 m	10.34 m

Table 10 -Closed loop RMSE by measurement-source configuration, all 34 segments.

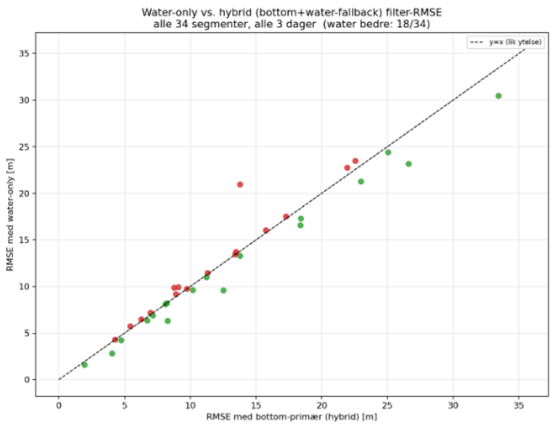


Figure 15 -Water-only vs. hybrid filter RMSE, all 34 segments (water better on 18 of 34).

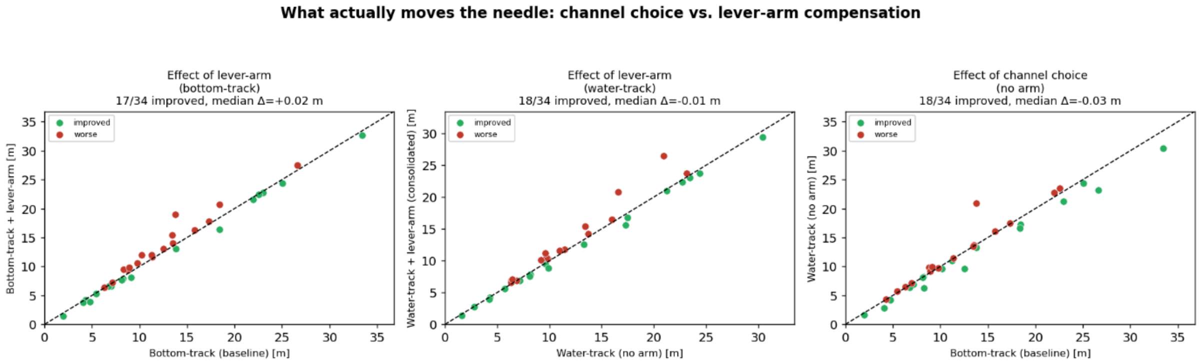
To verify that the behavior observed in the representative example was consistent across the full dataset, the same open-loop and closed-loop comparison was performed for all 34 recorded test segments.

The water-only solution produced a lower RMSE than the hybrid filter in 18 of the 34 test segments. The bottom-only filter was the least robust because it failed in the two segments where bottom lock was never established.

The results show that water velocity provides navigation performance comparable to bottom-track while offering significantly higher measurement availability. Although bottom-track remains the preferred measurement because it is independent of water currents, water velocity should be considered an additional navigation measurement rather than only a fallback during bottom-track dropouts.

8. Comparison of Navigation Configurations

8.1 Final Results



Figur 16 - Green points indicate test segments where the modified configuration achieved a lower RMSE than the baseline, while red points indicate segments where the RMSE increased.

Figure 16 compares the four navigation configurations evaluated in this project. The three scatter plots show the individual effects of lever-arm compensation and velocity-source selection, while the boxplot summarizes the RMSE distribution across all 34 test segments.

Changing the velocity measurement source from bottom-track to water-track had a larger influence on the navigation performance than applying lever-arm compensation. Lever-arm correction produced only small changes in whole-segment RMSE, which is consistent with the previous analysis showing that its benefit is mainly limited to turning maneuvers.

The consolidated configuration, combining water-track measurements with continuous lever-arm compensation, reduced the RMSE in 19 of the 34 evaluated test segments. Although the mean RMSE reduction across all segments was modest (approximately 0.5%), the consolidated configuration represents the most complete implementation developed during this project and combines the improvements investigated throughout the previous chapters.

Result	Value
Median RMSE improvement (consolidated vs. baseline)	≈ −1.6% (−2.6% excluding one outlier segment)
Best depth cell	2 m
Water-track coverage	100%
Bottom-track coverage	71.5–93.6%
Best single navigation source	Water-track (median RMSE 9.9 m vs. 10.7 m for bottom-track)
Lever-arm effect	up to 26% RMSE reduction in turning-heavy segments; negligible on average across all 34 segments
Filter consistency (NIS)	99.5% of updates within 95% bound (N = 47,848)
Closed-loop vs. open-loop	median RMSE 8.1 m vs. 75.2 m (≈ 9× improvement)

Table 11 -Summary of key quantitative results.

8. 2 Practical Challenges and Lessons Learned

This project involved considerably more than implementing a navigation filter. A significant part of the work focused on integrating the acoustic system on the Otter USV and establishing a reliable data acquisition pipeline. Since neither of us had experience with mechanical integration or acoustic instrumentation, substantial effort was spent designing the mounting system, routing cables, soldering connections and waterproofing the installation. These practical challenges became an important part of the project and provided valuable experience.

One of the main hardware challenges was integrating the Kongsberg CP300 with the Wideband Transceiver (WBT). After upgrading to EK80 version 0.90, the WBT was not consistently detected, requiring several network configuration changes before stable communication was achieved. During field testing, the EK80 occasionally lost bottom track, which could often be restored by disabling and re-enabling the bottom-track function.

The complete CP300 installation including the transducer, WBT, waterproof enclosure and an external 12 V battery increased the payload and shifted the vessel's centre of gravity, reducing stability. Because the WBT could not be powered reliably from the Otter platform, an external battery was required. Future work should investigate a fully integrated onboard power solution.

The project also demonstrated that while the CP300 provides accurate velocity measurements, its size, power requirements and installation complexity may limit its suitability for small USVs. Since Gudrun already carries a LiDAR sensor, future work could investigate LiDAR-based localisation as a lighter alternative for GPS-denied navigation in coastal environments.

Finally, the navigation filter is not completely independent of the integrated Seapath system because raw IMU measurements were not accessible. Although GPS measurements were excluded after initialization, future work should evaluate a standalone IMU together with additional independent sensors.

Overall, the project demonstrated that successful GPS-denied navigation depends not only on filter design but also on robust hardware integration, reliable communication and sensor calibration. The developed system provides a solid foundation for future work.

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